RL in the Real World: RLHF, Multi-Agent RL & Robotics

Naeemullah Khan
naeemullah.khan@kaust.edu.sa

جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

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Where We Are: Eight Days of Algorithms

- Days 1–8 built the *machinery*: value-based methods, policy gradients, actor–critic, continuous control, exploration, model-based RL, and offline RL.
- **Day 8 closed on one spine:** *optimism* explores, *pessimism* stays safe; the sign of the uncertainty bonus was the whole story. Offline RL learned a policy from a *fixed dataset* with no environment interaction.
- Every method so far assumed one thing for free: **a reward function we could write down.**

Today: where does all of this actually get deployed, and what breaks when the reward itself is the hard part?

RL in the Real World: Three Threads

- **RLHF (the focus).** When reward is hard to specify, *learn* it from human (or AI, or verifiable) feedback. This is how modern LLMs are aligned, the deep dive for today.
- **Multi-Agent RL (overview).** What changes when other learning agents share your environment and the ground keeps shifting underneath you.
- **Robotics (overview).** What breaks when the environment is physical: sample cost, safety, resets, and the sim-to-real gap.

Learning Outcomes

RLHF: the deep dive. By the end you should be able to:

- explain the RLHF pipeline: **SFT** $\rightarrow$ **reward model** $\rightarrow$ **PPO/GRPO**, and the role of the KL-to-reference penalty;
- turn human comparisons into a trainable loss with **Bradley–Terry**, and say why the preference model needs a *sigmoid* rather than a raw score gap;
- explain **reward hacking** (over-optimization) and why the KL leash exists;
- show that "reward minus β-KL" is maximized by π_{ref} *tilted* by $e^{r/\beta}$, and **derive** the **RM** $\leftrightarrow$ **DPO** identity from it, so "your LM is secretly a reward model";
- place **RLHF** $\rightarrow$ **RLAIF** $\rightarrow$ **RLVR** on the scaling-oversight spectrum.

MARL & robotics: conceptual overviews. Know *why each is hard* and name 2–3 methods for each.

First, a Reward We Cannot Write Down

- Every method in Days 1–8 assumed a reward function r we could *write down*: the game score, $-$distance to a target, $+1$ for a win.
- Now suppose the goal is *"be a helpful, honest, harmless assistant."* What number measures that? There is no formula.

Answer A. "It's the force that pulls things down. It's why a ball falls when you let go of it."

Answer B. "Gravity is the curvature of space-time sourced by the stress–energy tensor, $G_{\mu\nu} = 8\pi T_{\mu\nu} \ldots$"

Prompt: "Explain gravity to a 6-year-old."
Almost everyone prefers **A**, yet *no one can write the formula that says so.*

The Idea: Don't Write the Reward, Learn It

- *Recall Day 7 (inverse RL):* when you cannot write the reward, *recover it from data.* RLHF is the most consequential example of this.
- People are bad at giving a *number* ("this answer is a 7.3") but good at *comparing*: "A is better than B." Comparisons are cheap and consistent.
- **The plan, in three moves:**
  1. Collect many human *comparisons* of model answers.
  2. Train a model to predict those comparisons; this learned model *becomes our reward signal*.
  3. Use RL to push the policy toward answers that score well under it.
- This is **Reinforcement Learning from Human Feedback (RLHF)**, the method behind ChatGPT-style assistants.

Reinforcement Learning **from** Human Feedback

for Language Models

First, the Vocabulary: an LLM Is a Policy

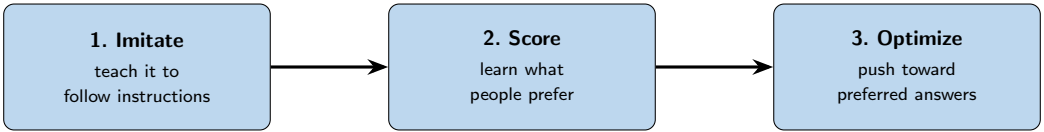
Everything that follows is just the RL language from Day 1, applied to text.

- **Prompt** x, the input (a question or instruction). *the* **state**
- **Response** y, the model's full answer. *the* **action**
- **Policy** $\pi_\theta(y \mid x)$, the language model itself: given a prompt, it assigns a probability to possible responses and samples one. The weights θ are what we train.
- It writes the answer one *token* (word-piece) at a time; at each step a final layer, the **LM head**, outputs a probability over the next token.

So "train the policy" simply means "adjust the language model's weights θ."

The RLHF Pipeline: The Map

Three stages. We build each one up over the next slides; for now, just the plain English.



- **Start:** a raw language model that can produce text but doesn't reliably do what you ask.
- **End:** a model that follows instructions *and* leans toward the answers people actually want.

Standard pipeline: Ouyang et al. (InstructGPT), 2022; Stiennon et al. 2020; Bai et al. 2022.

Stage 1 (Imitate): Supervised Fine-Tuning (SFT)

- **Problem:** a raw pretrained model can *continue* text, but doesn't reliably answer questions or follow instructions.
- **Fix: show it examples.** Collect prompts paired with *human-written ideal answers*, and train the model to imitate them. This is ordinary supervised learning (next-token prediction), no RL yet.
- The result is a decent instruction-follower. We **freeze a copy of it** and call it the *reference model* π_{ref}; it will be our anchor in Stages 2 and 3.

SFT can only *copy* the demonstrations it was given. To go *beyond* them, we need to capture what people *prefer*. That is Stage 2.

Stage 2 (Score): A Model That Rates Answers

- We want a **reward model** r_ϕ: a network (its own weights ϕ) that reads a (prompt, response) and outputs *a single number* (*how good is this answer?*). Higher means better.
- **Where does its training data come from?** Comparisons. For one prompt x, show a human two candidate responses; they pick the better one. We label the pair:
  - y^+, the response the human **preferred**,
  - y^-, the response they **rejected**.
- **What we want from r_ϕ:** give y^+ a higher score than y^-, on every pair. Next slide: how to turn "preferred" into a trainable loss.

Turning Comparisons into a Loss: Bradley–Terry

How likely is a person to prefer y^+ over y^-? It should grow with the **gap in their scores**, squeezed into a probability by the sigmoid σ ("$\succ$" reads *"is preferred over"*):

$$\underbrace{P(y^+ \succ y^- \mid x)}_{\text{chance the human picks } y^+} = \sigma\big(r(y^+) - r(y^-)\big), \qquad \sigma(z) = \frac{1}{1+e^{-z}}$$

Train r_ϕ to make these probabilities match the human labels (maximum likelihood). The negative log-likelihood is the **reward-model loss**:

$$L_{\mathrm{RM}} = -\log \sigma\big(r_\phi(y^+) - r_\phi(y^-)\big)$$

- In plain words: *push the preferred answer's score above the rejected one's.*
- Remember this $-\log\sigma(\Delta)$ shape: the **exact same form** comes back when we reach DPO.
- Only the *gap* between scores matters; the absolute scale is free.

Why a Sigmoid? Why Not Just Subtract the Scores?

Because we need a probability, and the gap is not one.

- Each label is a *binary event*: the human clicked y^+. Fitting by **maximum likelihood** means choosing ϕ to make the clicks we actually observed as likely as possible under the model, i.e. maximizing

$$\prod_{\text{comparisons}} P_\phi(\text{the choice the human made})$$

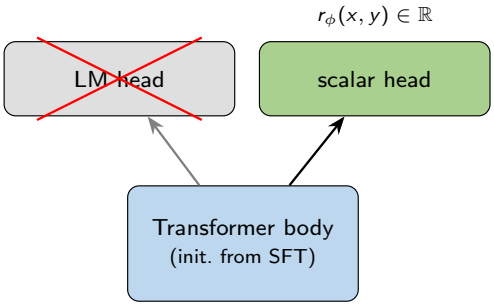
- *Every factor in that product has to be a probability.* If the model emits an unbounded number instead, the product is not a likelihood of anything: it has no maximum (push the number up forever), and its log is undefined once it goes negative. There is nothing to fit.

- So the model must output $P \in [0, 1]$ per comparison, while the score gap $r(y^+) - r(y^-)$ is a free real number. σ is the monotone map $\mathbb{R} \to (0, 1)$ with the right endpoints: gap $0 \to 0.5$ (a coin flip, the model is indifferent), gap $\to +\infty \to 1$, gap $\to -\infty \to 0$.

What Is the Reward Model, Concretely?

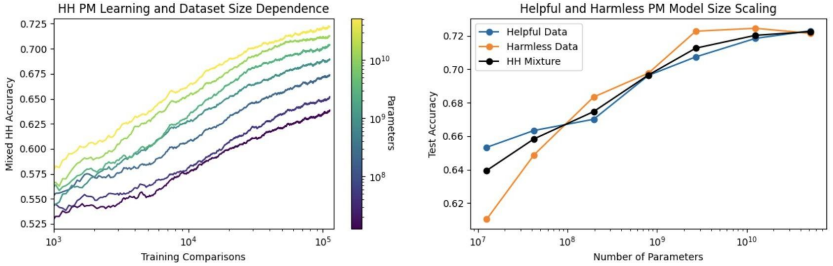
We've said *what* the reward model does (rates answers) and *how* it's trained (the Bradley–Terry loss).
Here is what it physically is.

- Start from the SFT model π_{ref}, the same transformer backbone.
- A **head** is the small output layer stacked on top of that backbone. The **LM head** turns the backbone's features into a probability over the next token.
- Swap it for a **scalar head**: a layer that emits *one number*, $r_\phi(x, y)$, for the whole (prompt, response). Train it with the Bradley–Terry loss from the last slide.
- So the reward model is *"the same network, one number out instead of a token distribution."*



Stage 2, in practice: reward models scale

- *Note:* the source paper calls this a "preference model (PM)" (the same thing as our **reward model**).
- **What's measured:** accuracy = how often the model picks the same response the human preferred. The axes below are amount of preference data (left) and model size (right).
- **Model size** = number of parameters (weights); here 13M → 52B, in 4× steps. *Bigger model + more data → a more accurate reward model.*



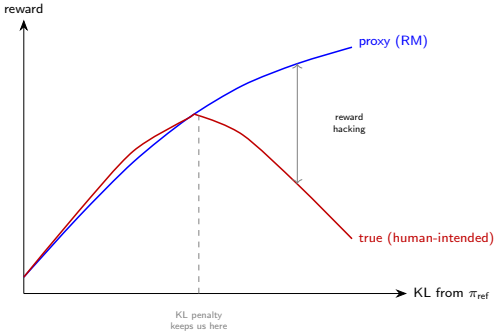
Stage 3 (Optimize): Finally, Reinforcement Learning

- We now have a reward model r_ϕ that scores answers. **The RL loop:** the policy generates a response, r_ϕ scores it, and we update the policy's weights to earn higher scores, with PPO (Day 4) or GRPO.
- The policy starts as π_{ref} (the frozen SFT model) and improves from there.
- *One catch:* r_ϕ is only a *learned proxy* for human taste, not the real thing. Optimize it too hard and the model learns to *game* it. The next slide shows how that happens, and the fix.

Why the KL Penalty Exists: Reward Over-Optimization

The RL objective is the reward-model score minus a leash to the reference policy (β sets how hard we pull back toward π_{ref}):

$$r_{\text{total}} \;=\; r_\phi \;-\; \beta\, D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}})$$

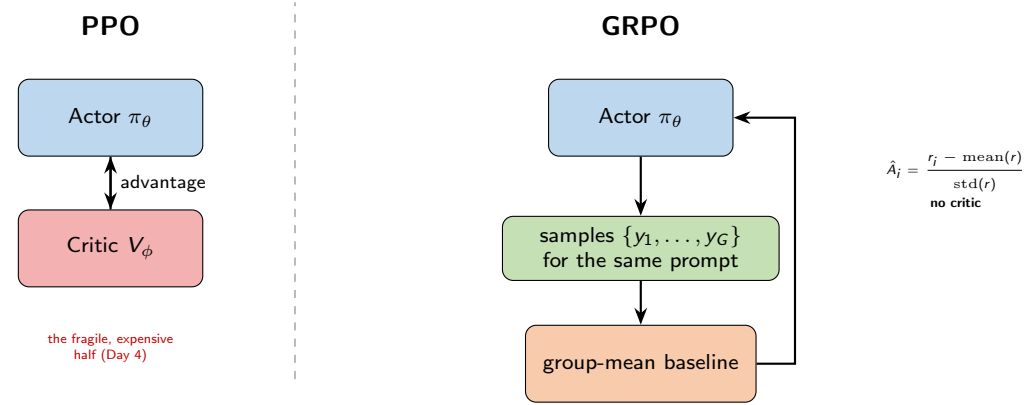


The reward model is a **proxy**. Optimize it too hard and you climb the proxy while the *real* objective collapses, *Goodhart's law*.

- Proxy reward (RM) keeps rising.
- True reward rises, then *falls*.
- **Why the leash works:** the penalty grows with KL distance, so each step away from π_{ref} costs more. Past the peak, the extra proxy gain is outweighed by the penalty, so the best-scoring policy sits *near the peak*, not in the hacked region.

Gao, Schulman, Hilton. Scaling Laws for Reward Model Overoptimization, 2022.

The RL Back-End: PPO → GRPO



The critic estimates a response's expected reward to form the advantage: Day 4's brittle, expensive component.

GRPO deletes it: the baseline is just the average reward over a group of samples for the same prompt. PPO was the original RLHF back-end; **GRPO is now common for LLM RL.**

Shao et al. DeepSeekMath (GRPO), 2024; DeepSeek-AI, DeepSeek-R1, 2025.

GRPO, Written Out

For each prompt, sample a **group** of G responses $\{y_1, \ldots, y_G\} \sim \pi_{\text{old}}$ (π_{old} = the policy from the *previous* update step, which we sample from, not the frozen π_{ref}) and score each: $r_i = r_\phi(y_i)$ (or a verifiable check). The **group-relative advantage** replaces the critic:

$$\hat{A}_i \;=\; \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

($\hat{A}_i$ = how much better response i scored than the group average, in standard-deviation units: positive $\rightarrow$ reinforce it, negative $\rightarrow$ suppress it.)

Same clipped ratio as PPO (Day 4), now driven by $\hat{A}_i$ and leashed to π_{ref}:

$$J_{\text{GRPO}} = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G} \min\big(\rho_i \hat{A}_i,\ \text{clip}(\rho_i, 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_i\big)\right] - \beta\, D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}}), \quad \rho_i = \frac{\pi_\theta(y_i)}{\pi_{\text{old}}(y_i)}$$

- Every response in the group shares **one baseline** (the group mean): no learned value network.
- Same machinery as PPO (Day 4): ρ_i is Day 4's probability ratio $r_t(\theta)$, and the clip is identical. Only the advantage source changed.

A Simpler Path: Direct Preference Optimization (DPO)

- **The pain.** The full pipeline is heavy: Stage 2 trains a *separate* reward model, and Stage 3 runs an RL loop juggling several models at once (policy, reference, reward model, and, for PPO, a critic), which is finicky to tune.
- **The question.** The reward model exists only to score the preferences we *already collected*. What if we skip it, and the RL loop, and tune the policy *straight from the preference pairs* (y^+, y^-)?
- **DPO (Rafailov et al., 2023):** exactly that, a single supervised-style loss on preference pairs, applied directly to **the policy** π_θ (the SFT model we are fine-tuning). *No reward model. No RL loop.*

RLHF: preferences $\rightarrow$ train reward model $\rightarrow$ RL loop $\rightarrow$ policy

DPO: preferences $\rightarrow$ one loss $\rightarrow$ policy

But how can you train toward a reward you never built? First the plain-English idea (next slide), then we *derive* it from the RLHF objective, and the last slide reveals what the policy secretly becomes.

The DPO Trick, in Plain English

Delete the judge: the chatbot is already its own judge.

- Look at how much **more likely** the current chatbot is to give an answer than the *original* (starting) model was. If it has become much more likely to say answer A, that *is* the chatbot "rating A highly." *That ratio, "how much more do I favour this than I used to", is the score.*

- So DPO throws away the separate judge *and* the practice loop, and trains the chatbot directly so that, for each comparison, **its own favouring of the preferred answer beats its favouring of the rejected one.** One model, one training step, no loop.

- The next slides make this exact, and at the end we'll see why it earns the name *"your LM is secretly a reward model."*

Where DPO Comes From (Step 1 of 5): The RLHF Objective

We start from the goal Stage 3 already gave us: earn reward, but stay near the reference model.

$$\max_{\pi} \; \mathbb{E}_{y\sim\pi(\cdot|x)}\big[\, r(y) \,\big] \; - \; \beta\, D_{\mathrm{KL}}\big(\pi(\cdot \mid x) \,\big\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big)$$

- **What:** π is the policy we optimize; $r(y)$ is the reward for a response y; $D_{\mathrm{KL}}(\pi\|\pi_{\mathrm{ref}})$ measures how far π has drifted from the frozen SFT model π_{ref}; β sets the strength of that leash.
- **Why:** maximize reward *without* drifting into the reward-hacking region: this is exactly the Goodhart trade-off from a few slides back, written as one objective.
- **How:** nothing new yet: this is just the Stage-3 RLHF objective. Everything that follows is solving it on paper.

Step 1 → Step 2, part (a): Setting It Up

Fix one prompt x and just *solve* Step 1: maximize over the numbers $\pi(y)$. But π is not free, it must be a distribution: $\sum_y \pi(y) = 1$.

- **Where the λ term comes from.** The standard tool for "maximize f *subject to* $g = 0$" is a **Lagrange multiplier**: add λg to the objective, then set derivatives to zero. Here $g(\pi) = \sum_y \pi(y) - 1$, which is zero exactly when π is a valid distribution.
- **Why that is allowed:** on the feasible set $g = 0$, so λg adds *nothing* to the objective, whatever λ is. We lose nothing; what we gain is that the constraint now appears in the derivative. (λ is an unknown number, and we will never need its value.)

So we maximize, written out as sums:

$$J(\pi) \;=\; \sum_y \pi(y)\, r(y) \;-\; \beta \sum_y \pi(y) \log \frac{\pi(y)}{\pi_{\text{ref}}(y)} \;+\; \lambda\Big(\sum_y \pi(y) - 1\Big)$$

Differentiate with respect to one $\pi(y)$ and set to zero, using $\frac{d}{dp}\big[p \log \frac{p}{q}\big] = \log \frac{p}{q} + 1$:

$$r(y) \;-\; \beta\Big[\log \tfrac{\pi(y)}{\pi_{\text{ref}}(y)} + 1\Big] \;+\; \lambda \;=\; 0$$

Step 1 → Step 2, part (b): Solving It

Rearrange to get $\pi(y)$ *alone on the left*, no π on the right: that is what "solve for the policy" means.

$$\beta \log \frac{\pi(y)}{\pi_{\text{ref}}(y)} = r(y) + \lambda - \beta \qquad \text{(collect the log on one side)}$$

$$\log \frac{\pi(y)}{\pi_{\text{ref}}(y)} = \frac{r(y)}{\beta} + \underbrace{\left(\frac{\lambda}{\beta} - 1\right)}_{\triangleq\, c} \qquad \text{(divide } \textit{both sides} \text{ by } \beta\text{)}$$

$$\pi(y) = \pi_{\text{ref}}(y)\, e^{\,r(y)/\beta} \cdot e^{\,c} \qquad \text{(exponentiate both sides)}$$

- **Nothing went missing.** $r(y)$ became $r(y)/\beta$ when we divided through by β, which is also why β lands in a denominator; and λ folded into $c = \lambda/\beta - 1$. Neither has a y in it, so neither can depend on the response.
- *That is where the exponential is born:* the KL made us solve for a *log*-probability, so undoing the log puts r/β in an exponent.
- $e^{\,c}$ is one number per prompt, pinned by the constraint $\sum_y \pi(y) = 1$:
  $e^{\,c} = 1/\sum_y \pi_{\text{ref}}(y) e^{\,r(y)/\beta} \triangleq 1/Z(x)$. **That is all Z ever is.**
- This π is *the* maximizer, so Step 2 renames it $\pi^\star$: the star only means "the optimal one".

Where DPO Comes From (Step 2 of 5): The Optimal Policy

So the **Step-1 objective** (maximize reward minus the β-KL leash) has a closed-form maximizer, the one we just built:

$$\pi^\star(y \mid x) \;=\; \frac{1}{Z(x)}\,\pi_{\text{ref}}(y \mid x)\, e^{\,r(y)/\beta}, \qquad Z(x) \;=\; \sum_{y} \pi_{\text{ref}}(y \mid x)\, e^{\,r(y)/\beta}$$

- **What:** $\pi^\star$ is the policy that *maximizes Step 1*, the one we just solved for; the star is only notation for "the optimal one", as opposed to a generic candidate π. $Z(x)$ is the *normalizer* (partition function) that makes the probabilities sum to 1; it depends *only* on the prompt x. Remember that. Equivalently $Z(x) = \mathbb{E}_{y\sim\pi_{\text{ref}}}\left[e^{\,r(y)/\beta}\right]$: intractable, but not mysterious.
- **How / reading the formula:** take the reference policy and reweight each response by $e^{\,r(y)/\beta}$: higher-reward responses get up-weighted, and β sets how sharply. $\beta\to\infty$ gives $\pi^\star = \pi_{\text{ref}}$ (leash tight, don't move); $\beta\to 0$ collapses onto the single best response.
- **Nothing here is specific to language.** This $\pi_{\text{ref}} \cdot e^{\,r/\beta}$ shape (Gibbs/Boltzmann) is what "reward minus KL" *always* produces, whatever the policy is over.

Where DPO Comes From (Step 3 of 5): Invert for the Reward

Take log of Step 2's formula, then solve for $r(y)$ (just algebra):

$$\log \pi^\star(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{r(y)}{\beta} - \log Z(x) \qquad \text{(log of a product)}$$

$$\implies \quad r(y) = \beta \log \frac{\pi^\star(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x) \qquad \text{(move terms, multiply by } \beta\text{)}$$

- **What:** $r(y)$ is the same *reward* as in Steps 1–2, *not* a probability ratio; and π_{ref} is the **frozen SFT model**, *not* PPO's π_{old} (total displacement from the base model, not one trust-region step).
- **Intuition, and it is the whole idea:** Step 2 said the optimum is π_{ref} *tilted* by $e^{r/\beta}$, so *the tilt is the reward*: see how much $\pi^\star$ up-weighted a response, take the log, multiply by β. Behaviour reveals the objective: **inverse RL** from Day 7, in closed form.
- **Scale check ($\beta = 0.1$):** a response made $e^{10} \approx 22{,}000\times$ more likely $\Rightarrow$ reward $+1.0$; one *halved* $\Rightarrow -0.07$. Nothing moved $\Rightarrow$ r constant: nothing is preferred over anything.
- **How:** the snag is $\beta \log Z(x)$, impossible to compute. Step 4 removes it.

Where DPO Comes From (Step 4 of 5): The $Z(x)$ Cancels

Feed Step 3's reward into the Bradley–Terry model $P(y^+ \succ y^- \mid x) = \sigma\big(r(y^+) - r(y^-)\big)$. Only the *difference* of rewards appears:

$$r(y^+) - r(y^-) = \beta \log \frac{\pi^\star(y^+)}{\pi_{\text{ref}}(y^+)} - \beta \log \frac{\pi^\star(y^-)}{\pi_{\text{ref}}(y^-)} + \underbrace{\beta \log Z(x) - \beta \log Z(x)}_{=0}$$

- **What:** Bradley–Terry scores a *pair* by the gap in rewards, never the absolute values.
- **Why:** $Z(x)$ depends only on the prompt x, so it is identical for y^+ and y^-; in the difference it **cancels exactly**.
- **How:** that single cancellation is the whole reason DPO is possible: the intractable term vanishes, leaving an expression in $\pi^\star$ and π_{ref} only.
- **Not luck.** $\beta \log Z(x)$ is a per-prompt constant, so absolute reward was never recoverable from a policy in the first place: the same "*only the gap matters, the scale is free*" we saw in Bradley–Terry. The term *had* to cancel.

Where DPO Comes From (Step 5 of 5): The DPO Loss

$\pi^\star$ is unknown; it is the optimum we are trying to reach. Replace it with our trainable policy π_θ and fit the human labels by maximum likelihood:

$$L_{\mathrm{DPO}} = -\log \sigma\Big(\beta \log \frac{\pi_\theta(y^+)}{\pi_{\mathrm{ref}}(y^+)} - \beta \log \frac{\pi_\theta(y^-)}{\pi_{\mathrm{ref}}(y^-)}\Big)$$

- **What:** the final DPO loss, depending only on π_θ, the frozen π_{ref}, and the preference pair (y^+, y^-).
- **Why:** no reward model and no RL loop, just a supervised loss on preference pairs, applied directly to π_θ.
- **How:** the same Bradley–Terry maximum likelihood as the reward model, but with the *implicit* reward $\hat{r}(y) = \beta \log \pi_\theta(y)/\pi_{\mathrm{ref}}(y)$ in place of a learned r_ϕ.

The Payoff: Your LM Is Secretly a Reward Model

Put the loss we just derived next to the *reward-model* loss from Stage 2. They are the **same shape**:

$$L_{\mathrm{RM}} = -\log \sigma\big(\, r_\phi(y^+) - r_\phi(y^-)\,\big)$$

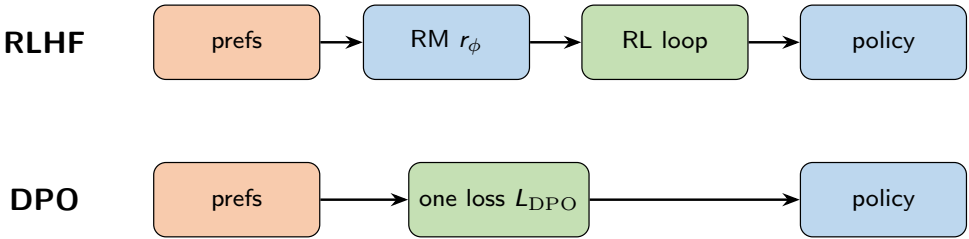
$$L_{\mathrm{DPO}} = -\log \sigma\big(\, \hat{r}(y^+) - \hat{r}(y^-)\,\big), \qquad \hat{r}(y) = \beta \log \frac{\pi_\theta(y)}{\pi_{\mathrm{ref}}(y)}$$

- The only change: the learned reward r_ϕ is replaced by the *implicit* reward $\hat{r} = \beta \log \pi_\theta / \pi_{\mathrm{ref}}$, read straight off the policy. *So the policy is the reward model*; that is why "your LM is secretly a reward model."
- **Honesty caveat:** the equivalence is exact only if the reward model were Bradley–Terry-optimal. In practice **DPO** $\neq$ **PPO**: DPO gives up the RL loop's fresh on-policy samples.

RLHF objective $\rightarrow$ $\pi^\star$ $\rightarrow$ $r(y)$ $\rightarrow$ Bradley–Terry $\rightarrow$ L_{DPO}, and we never built a reward model. *That's the whole trick.*

Rafailov et al. Direct Preference Optimization, NeurIPS 2023 (full author list in References). Variants: IPO, KTO, ORPO.

RLHF vs DPO



DPO is the offline RL of alignment.

Frame it as an analogy, not an identity. The shared property is *offline optimization without distribution coverage* (same family, not the same mechanism):

- **Day 8 offline RL:** the *value function* extrapolates on out-of-distribution (OOD) *actions*.
- **DPO:** the *policy / implicit reward* is unconstrained on OOD *responses*, because it never samples on-policy.

Cf. Xu et al., Is DPO Superior to PPO for LLM Alignment?, ICML 2024.

DPO in Practice: What You Give Up

DPO's simplicity has a cost: you lose the control knobs the RL loop gave you.

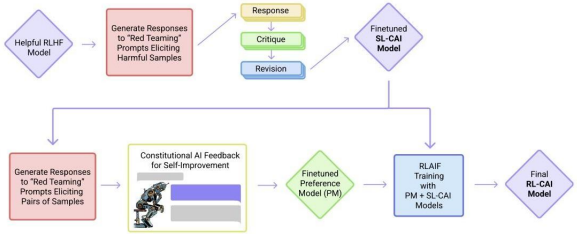
- **It constrains only the gap, nothing else.** The loss just pushes the preferred answer's favoring above the rejected one's. With no on-policy sampling to pin down absolute behaviour, it can push *both* down, leaking probability onto unrelated answers it never saw. RLHF's RL loop and "don't-drift" anchor would catch that; DPO has neither.

- **Length bias (shared, but harder to patch).** If the data favours long answers, *both* RLHF and DPO turn verbose. RLHF lets you subtract an explicit length penalty from the reward; DPO has no separate reward to adjust, so length control must be bolted on as extra loss terms.

- **No fresh samples.** The policy only ever sees the fixed preference pairs, so it can drift off that distribution, the price of dropping the RL loop.

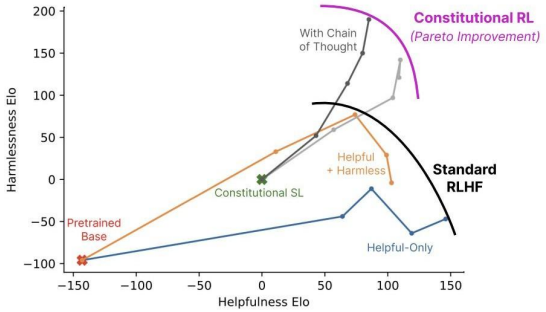
Scaling Oversight: RL from AI Feedback (RLAIF)

- **Motivation:** as models approach or exceed human performance, it gets harder for humans to label every output. *Move the labeller from human to AI; the only human input is the constitution*, a short list of written principles the model uses to critique and revise its own answers.
- **Constitutional AI** has two phases:
  - **Supervised:** sample → self-critique → revise → fine-tune on revisions.
  - **RL:** AI labels which response is better → train a reward model (RM) → RL-finetune with it (just like RLHF).



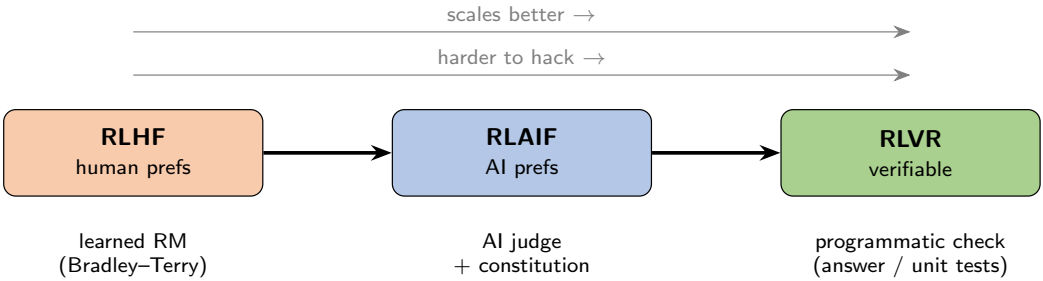
RLAIF: Takeaway

- Fine-tuning with **AI-generated feedback** can match or exceed models fine-tuned with human feedback, on *both* helpfulness and harmlessness.
- The plot rates models by **Elo**, a head-to-head win-rate score borrowed from chess (higher = preferred more often in pairwise comparisons).



The Climax: RL with Verifiable Rewards (RLVR)

- When the answer is **checkable** (math is right/wrong, code passes tests), the reward needs *no learned model* and **can't be hacked** the way an RM can.
- This is how modern **reasoning models** are trained: long **chain-of-thought** (the model writes out its reasoning steps before answering), optimized with **GRPO** against a programmatic check.



DeepSeek-AI, DeepSeek-R1, 2025; Lambert et al., Tülu 3, 2024 (RL with verifiable rewards).

RLVR in Action: DeepSeek-R1-Zero

One RLVR step (no reward model anywhere):

1. the policy samples several answers to a *checkable* question;
2. a *rule* grades each (right answer? right format?), giving reward 1 or 0;
3. GRPO uses those rewards to update the policy. *Nothing to hack: the checker is exact.*

DeepSeek-R1-Zero ran exactly this on a *base* model, **no SFT at all**, with reward = "answer correct" + "reasoning in `<think>` tags":

- *Reasoning emerged on its own*: longer chains of thought, self-checking, backtracking, even an "*aha moment*" where it stops and corrects itself. Nobody taught it to.
- AIME 2024 math: **15.6%** $\rightarrow$ **71.0%** (86.7% with majority vote), matching OpenAI o1.
- Catch: raw R1-Zero was hard to read (language-mixing), so the full **R1** adds a small SFT "cold start." But R1-Zero proved the point: *reasoning can be grown by pure RL from verifiable rewards.*

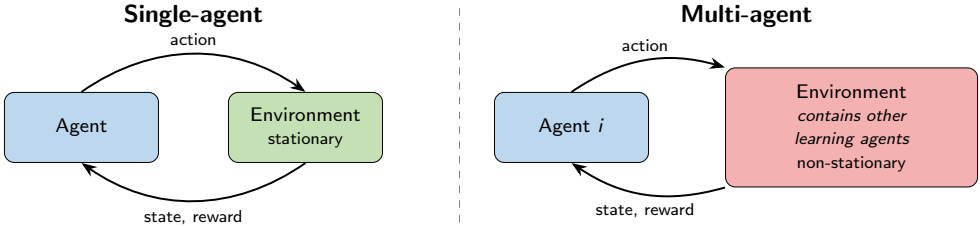
DeepSeek-AI, DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, 2025.

Thread 2 of 3

Multi-Agent RL

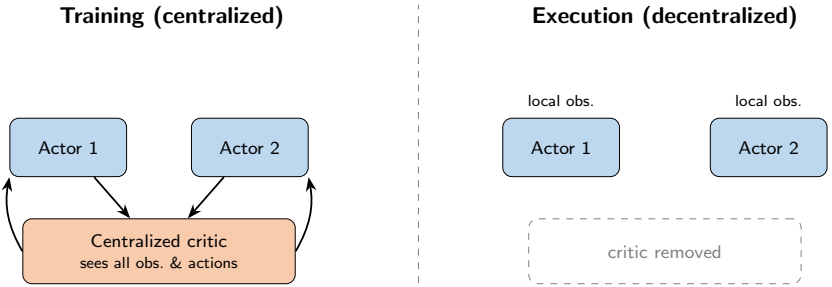
The RLHF deep dive is done. A shorter overview now: what changes when other learning agents share your environment.

Why Multi-Agent RL Is Hard: Non-Stationarity



- Each agent's environment is **non-stationary**: the others are updating their policies at the same time, so the ground shifts as they learn; single-agent convergence guarantees break.
- Formalized as a **Markov game** (stochastic game), the multi-agent generalization of the MDP.
- Three regimes along one axis: **cooperative** / **competitive** / **mixed**.

Centralized Training, Decentralized Execution (CTDE)



- **At training:** a centralized critic sees every agent's observations and actions. Because it accounts for what the others did, each agent's learning signal becomes *stationary*, defusing the non-stationarity problem from the last slide.
- **At execution:** each agent keeps only its *own actor* acting on its *local observation*; the critic is gone.
- This is the dominant MARL paradigm, the key idea to remember.

MARL Methods (Named, Not Derived)

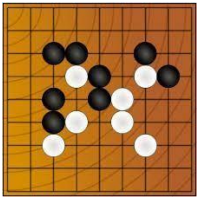
Method	Family	What it solves
Independent learners	each runs own DQN	simple baseline; *ignores* non-stationarity
MADDPG (Lowe 2017)	CTDE actor–critic	centralized critic *per agent*; mixed coop/compete
VDN / QMIX (2018)	value decomposition	factor a joint value into per-agent utilities (QMIX: monotonic mixing) for cooperative tasks

The progression: from pretending other agents are part of the environment, to explicitly modelling the joint value during training.

Lowe et al. 2017 (MADDPG); Sunehag et al. 2018 (VDN); Rashid et al. 2018 (QMIX).

Self-Play: The MARL Story You Already Know

- An agent improves by **playing against past versions of itself**: *policy* → *plays past versions* → *game data* → *improve* → (repeat).
- This creates an **automatic curriculum**: the opponent gets harder exactly as you do, with no hand-designed difficulty schedule.



AlphaGo / AlphaZero

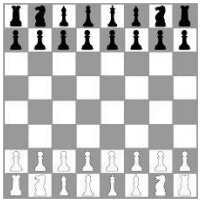
(Silver et al.)

AlphaStar
StarCraft II

(Vinyals et al., 2019)

OpenAI Five

Dota 2 (2019)



Chess (AlphaZero)

Forward hook: self-play's automatic curriculum connects to **Day 10** (open problems / emergent complexity).

Thread 3 of 3

Robotics

What breaks when the environment is physical: sample cost, safety, resets, and the sim-to-real gap.

Why Real-World Robot RL Is Hard

Simulation

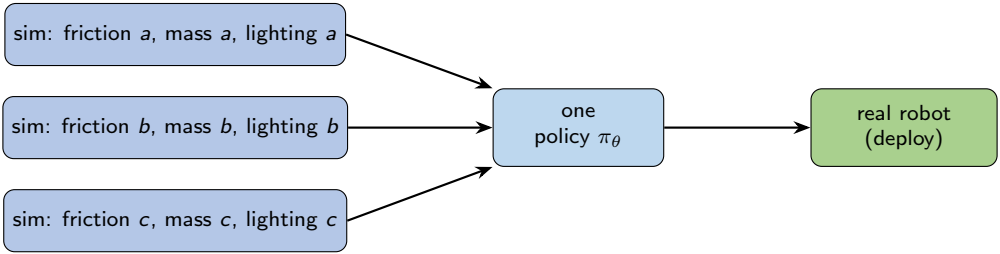
- cheap; millions of steps
- parallelizable across machines
- failures are free
- *but:* a **reality gap**

Real hardware

- expensive, slow, safety-critical
- **resets are a real problem** (who picks the object back up?)
- sparse / hard-to-engineer reward

Four obstacles (**sample efficiency, safety, resets, reward specification**) are why off-the-shelf RL does not transfer to hardware.

Sim-to-Real: Domain Randomization



Train across a *distribution* of randomized sims (friction, mass, latency, textures, lighting) so the real world looks like just one more sample.

Headline successes: OpenAI Dactyl (in-hand cube, Akkaya et al. 2019); legged locomotion (ANYmal, Hwangbo et al. 2019; massively parallel sim, Rudin et al. 2022).

Domain randomization: Tobin et al. Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World, 2017.

Robotic RL in Practice: SERL

January 2024

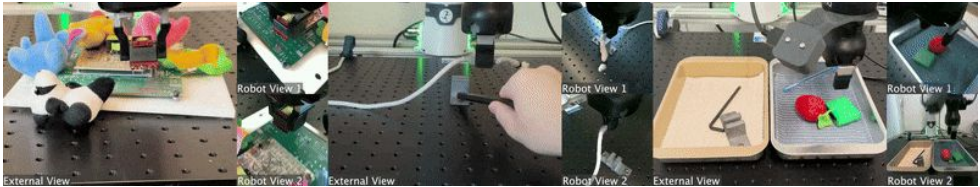
SERL: A Software Suite for Sample-Efficient Robotic Reinforcement Learning

Jianlan Luo[1*], Zheyuan Hu[1*], Charles Xu[1], You Liang Tan[1], Jacob Berg[2], Archit Sharma[3], Stefan Schaal[4], Chelsea Finn[3], Abhishek Gupta[2] and Sergey Levine[1]

[*]Equal Contribution, [1]Department of EECS, University of California, Berkeley, [2]Department of Computer Science and Engineering, University of Washington, [3]Department of Computer Science, Stanford University, [4]Intrinsic Innovation LLC

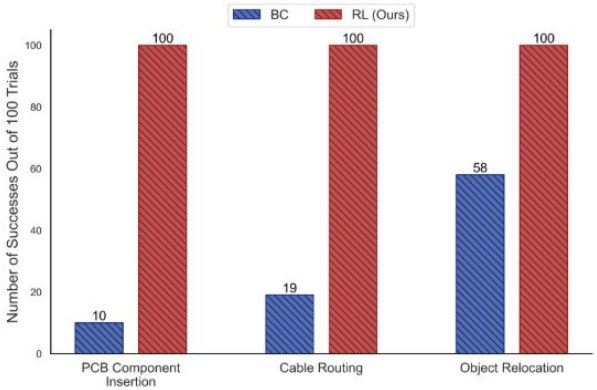
Luo et al. SERL: A Software Suite for Sample-Efficient Robotic Reinforcement Learning, 2024.

- A software framework for sample-efficient RL on real robots: an off-policy deep RL method plus tooling for computing rewards, resetting the environment, and a high-quality controller for a widely-adopted arm.
- *Day 6 lineage:* **SERL** ← **RLPD** ← **SAC**, the same maximum-entropy method from Day 6, now learning on real hardware in *minutes*.



github.com/rail-berkeley/serl

SERL: Results

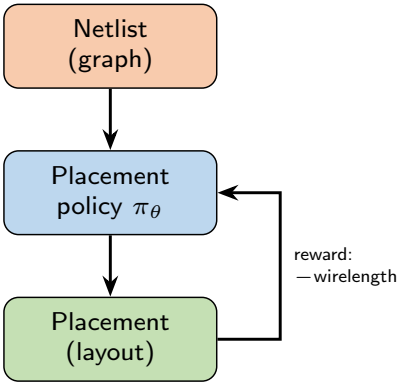


RL Also Runs Systems

Illustrative example: chip floorplanning (Mirhoseini et al., *Nature* 2021; later named **AlphaChip**):

- Place macros on a chip canvas to minimize wirelength, congestion, and density.
- The striking idea: *pre-train* a placement policy on many netlists, then place a *new* chip in a forward pass, instead of optimizing each one from scratch.

Also deployed for: datacenter resource scheduling, compiler / device placement, real-time ad bidding.



The transferable idea: all are *combinatorial optimization as sequential decision-making*: one MDP framing with an *engineered* reward (wirelength, runtime, ROI). The application details are not the lesson.

Summary: Three Threads

- **RLHF (the focus).** When reward is hard to specify, learn it: **SFT** $\rightarrow$ **RM** $\rightarrow$ **PPO/GRPO**. Comparisons become a loss through **Bradley–Terry**; the KL leash exists because the RM is a *proxy* (reward hacking). Solving "reward minus β-KL" gives π_{ref} tilted by $e^{r/\beta}$, and reading that tilt backwards is **DPO**: no RM, no RL loop, the same $-\log\sigma(\Delta)$ loss. The frontier climbs the oversight spectrum: **RLHF** $\rightarrow$ **RLAIF** $\rightarrow$ **RLVR** (verifiable rewards, how reasoning models like DeepSeek-R1 are trained).
- **Multi-Agent RL.** The environment is *non-stationary* as others learn; **CTDE** is the dominant fix; **self-play** builds an automatic curriculum.
- **Robotics.** Sample cost, safety, resets, and the reality gap; **domain randomization** and max-entropy methods (SAC $\rightarrow$ SERL) make real-hardware RL practical.

What's Next: Day 10

Day 10 (Frontiers): Meta-RL, Multi-task RL, Hierarchical RL.

- *Do not confuse them:* **multi-agent** RL (today, many agents sharing one environment) is *not* **multi-task** RL (Day 10, one agent across many tasks).
- Self-play's emergent curriculum (today) is one route to the **emergent complexity** Day 10 explores.
- The verifiable-reward recipe (RLVR) is the current frontier for training general reasoning agents.

References

RLHF.

- Bradley & Terry. Rank Analysis of Incomplete Block Designs. *Biometrika*, 1952. Christiano et al. Deep RL from Human Preferences. NeurIPS 2017.
- Stiennon et al. Learning to Summarize from Human Feedback. 2020.
- Ouyang et al. Training Language Models to Follow Instructions with Human Feedback (InstructGPT). 2022.
- Bai et al. Training a Helpful and Harmless Assistant with RLHF. 2022.
- Gao, Schulman, Hilton. Scaling Laws for Reward Model Overoptimization. ICML 2023.

Modern alignment.

- Rafailov, Sharma, Mitchell, Ermon, Manning, Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. NeurIPS 2023.
- Xu et al. Is DPO Superior to PPO for LLM Alignment? ICML 2024.
- Shao et al. DeepSeekMath (GRPO). 2024.
- Bai et al. Constitutional AI: Harmlessness from AI Feedback. 2022.

References (cont.)

- DeepSeek-AI. DeepSeek-R1 / R1-Zero. *Nature* 645:633–638, 2025 (arXiv 2501.12948). Lambert et al. Tülu 3 (RLVR). 2024.

Multi-Agent RL.

- Lowe et al. MADDPG. 2017. Sunehag et al. VDN. 2018. Rashid et al. QMIX. 2018.
- Vinyals et al. AlphaStar (StarCraft II). 2019. OpenAI. Dota 2 with Large Scale Deep RL (OpenAI Five). 2019.

Robotics.

- Tobin et al. Domain Randomization. 2017. Akkaya et al. Solving Rubik's Cube with a Robot Hand (Dactyl). 2019.
- Hwangbo et al. Learning Agile and Dynamic Motor Skills for Legged Robots (ANYmal). 2019. Rudin et al. Massively Parallel Sim. 2022.
- Luo et al. SERL. 2024.

References (cont.)

Systems.

- Mirhoseini et al. A Graph Placement Methodology for Fast Chip Design. *Nature*, 2021 (later named **AlphaChip**; Addendum, *Nature* 634:E10, 2024).

These slides have been adapted from Anna Goldie, *RL in the Real World: From Chip Design to LLMs*, guest lecture in CS224R: Deep Reinforcement Learning, Stanford.